Software Engineering

Assignment-10

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

For each of the following questions one or more of the given options are correct. Choose the correct options.

QUESTION 1:

A software organization observes from past projects that approximately 40% of defects are detected during unit testing, 35% during integration testing, and 25% during system testing. If the total planned testing effort for a project is 200 person-months, how much effort should ideally be allocated to integration testing?

a. 50 person-months
b. 60 person-months
c. 70 person-months
d. 80 person-months
e. 90 person-months

Correct Answer: c. 70 person-months

Detailed Solution:

Integration testing effort = $35\% \times 200 = 70$person-months.

QUESTION 2:

Consider a function named triangleType(a,b,c) that determines whether three integer side lengths form an Equilateral, Isosceles, or Scalene triangle. Assuming only valid equivalence classes are considered, what is the minimum number of test cases required for equivalence class testing?

a. 1
b. 2
c. 3
d. 4
e. 5

Correct Answer: c. 3

Detailed Solution:

The valid equivalence classes are Equilateral, Isosceles, and Scalene triangles. One test case is needed for each class.

QUESTION 3:

Which of the following are not black-box testing techniques?

a. Equivalence class testing
b. Boundary value analysis
c. Statement coverage testing
d. Path testing
e. Decision table testing

Correct Answer: c. Statement coverage testing, **d**. Path testing

Detailed Solution:

Statement coverage and path testing are white-box testing techniques because they depend on the internal structure and control flow of the program.

QUESTION 4:

Assume that a web-based examination system has three input fields: Subject, Difficulty Level, and Question Type.

- Subject can take 5 valid values: {Math, Physics, Chemistry, Biology, English}
- Difficulty Level can take 3 valid values: {Easy, Medium, Hard}
- Question Type can take 4 valid values: {MCQ, MSQ, NAT, Essay}

Assuming that the interface allows only valid inputs, how many test cases are required for exhaustive functional testing?

a. 12
b. 48
c. 60
d. 75
e. 120

Correct Answer: c. 60

Detailed Solution:

Exhaustive testing requires testing all possible valid combinations: $5 \times 3 \times 4 = 60$test cases.

QUESTION 5:

Which of the following statements are false regarding white-box testing?

a. White-box testing requires knowledge of the program's internal logic and structure
b. Path testing is a white-box testing technique
c. White-box testing is also known as behavioral testing
d. Statement coverage is one possible white-box testing strategy
e. Test cases in white-box testing are designed independently of the program code

Correct Answer: c. White-box testing is also known as behavioral testing, **e.** Test cases in white-box testing are designed independently of the program code

Detailed Solution:

White-box testing is structure-based testing, not behavioral testing. Its test cases are designed using knowledge of the program code and logic.

QUESTION 6:

Consider the function checkTriangle(float a, float b, float c) that determines whether three side lengths can form a valid triangle. For equivalence class testing, at the first level of the equivalence class hierarchy, valid and invalid equivalence classes can be identified. The valid set of input values can be further divided into how many equivalence classes?

a. 1
b. 2
c. 3
d. 4
e. 5

Correct Answer: c. 3

Detailed Solution:

The valid triangle inputs can be divided into three equivalence classes: Equilateral, Isosceles, and Scalene triangles.

QUESTION 7:

Which one of the following assumptions underlies boundary value analysis testing?

a. Errors are more likely to occur near the extreme ends of input ranges
b. All invalid inputs produce identical program behavior
c. Programs behave identically for all valid inputs
d. Boundary values are always invalid inputs
e. Every software fault can be detected using boundary values alone

Correct Answer: a. Errors are more likely to occur near the extreme ends of input ranges

Detailed Solution:

Boundary Value Analysis assumes that errors are more likely to occur near the boundaries or extreme ends of input ranges.

QUESTION 8:

Suppose a function named check(a, b, c) takes three Boolean variables as input parameters. How many valid equivalence classes can be designed for black-box testing of the function?

a. 4
b. 6
c. 8
d. 9
e. 12

Correct Answer: c. 8

Detailed Solution:

Each Boolean variable has 2 possible values (True/False). For 3 variables, total valid combinations are $2^3 = 8$.

QUESTION 9:

Which of the following statements regarding unit testing, drivers, and stubs are correct?

a. Unit testing is performed after integrating all modules of the software
b. A driver simulates the behavior of a calling function and provides input data to the unit under test
c. A stub simulates the behavior of a module that has not yet been implemented
d. Unit testing focuses on testing individual methods, modules, classes, or components in isolation
e. Drivers are mainly used to replace lower-level modules called by the tested component
f. Stubs are commonly used when the called modules are unavailable during testing

Correct Answer: b. A driver simulates the behavior of a calling function and provides input data to the unit under test, **c.** A stub simulates the behavior of a module that has not yet been implemented , **d.** Unit testing focuses on testing individual methods, modules, classes, or components in isolation, **f.** Stubs are commonly used when the called modules are unavailable during testing

Detailed Solution:

Drivers provide input to the unit under test, while stubs simulate unavailable called modules. Unit testing tests individual components in isolation.

QUESTION 10:

Consider a function substr(s1, s2) that takes two strings where s1 can have a maximum length of 20 characters and s2 can have a maximum length of 5 characters. The function checks whether s2 is a substring of s1. Which of the following represent valid equivalence classes for testing?

a. s1 length between 1 and 20 characters
b. s1 length greater than 20 characters
c. s2 is a substring of s1
d. s2 is not a substring of s1
e. s2 length exactly equal to 3 only
f. s1 length always equal to 20 only

Correct Answer: a. s1 length between 1 and 20 characters, **b**. s1 length greater than 20 characters, **c**. s2 is a substring of s1, **d**. s2 is not a substring of s1

Detailed Solution:

Equivalence classes include valid/invalid string lengths and cases where s2 is or is not a substring of s1. Single exact lengths are not equivalence classes.
